ECE 305 Homework SOLUTIONS: Week 11

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1) In an MOS capacitor, **everything depends on the bandbending in the semiconductor**. If ϕ_S is the potential at the surface, and $\phi = 0$ is the potential in the bulk, then $-q\phi_S$ is the total bandbending in the semiconductor. A negative ϕ_S means the bands bend up, and a positive ϕ_S means the bands bend down. This question helps familiarize you with surface potential and bandbending. Assume a Si MOS capacitor at room temperature.

1a) Assume $N_A = 10^{17}$ cm^{-3} and compute $E_i - E_F$ and the related potential, $\phi_F = \left(E_i - E_F\right)/q$, which plays an important role in MOS electrostatics.

Solution:
To determine $E_i - E_F$, we begin with the relation between E_i and E_F:

$$p_0 = N_A = n_i e^{\left(E_i - E_F\right)/k_B T}$$

and solve it to find

$$\left(E_i - E_F\right) = k_B T \ln\left(\frac{N_A}{n_i}\right)$$

The potential is

$$\phi_F = \frac{\left(E_i - E_F\right)}{q} = \frac{k_B T}{q} \ln\left(\frac{N_A}{n_i}\right)$$

Putting in numbers:

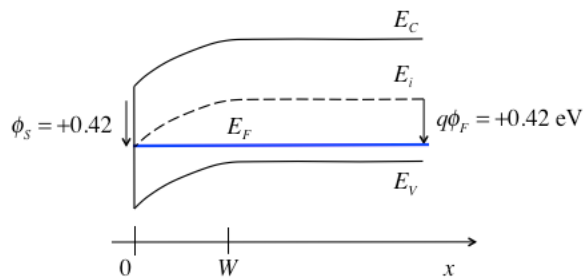
$$\phi_F = 0.026 \ln\left(\frac{10^{17}}{10^{10}}\right) = 0.42 \text{ V} \qquad \boxed{\phi_F = +0.42 \text{ V}}$$

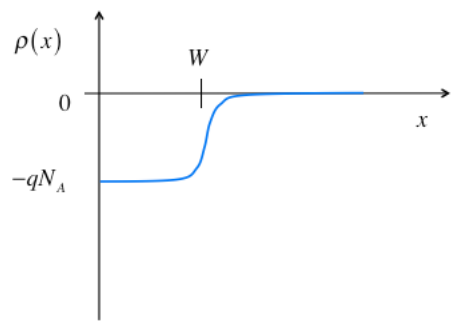
Note that ϕ_F is positive in a P-type semiconductor (and negative in an N-type semiconductor).

1b) Assume $\phi_S = \phi_F$ and sketch the energy band diagram and the charge density, $\rho(x)$ vs. position in the semiconductor.

HW11 Solutions (continued):

Solution:

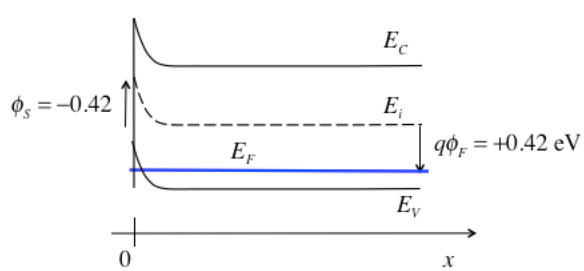




Note: When solving for the electric fields in the depletion approximation, we approximate the charge density as a rectangle with an abrupt edge at $x = W$.

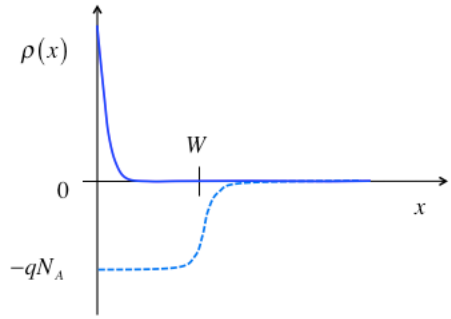
1c) Assume $\phi_S = -\phi_F$ and sketch the energy band diagram and the charge density, $\rho(x)$ vs. position in the semiconductor.

Solution:



Negative surface potential means that the bands bend up at the surface. Majority carrier holes will pile up at the surface and that there is NO depletion region.

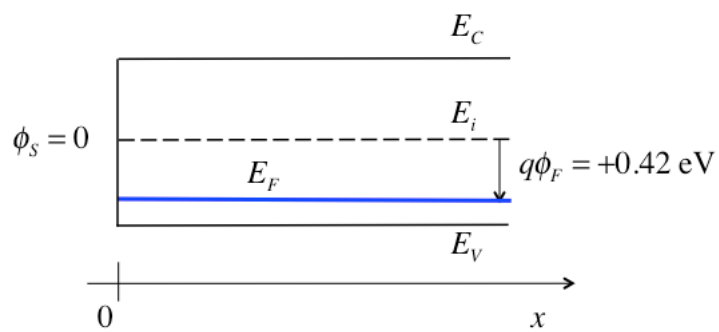
HW11 Solutions (continued):



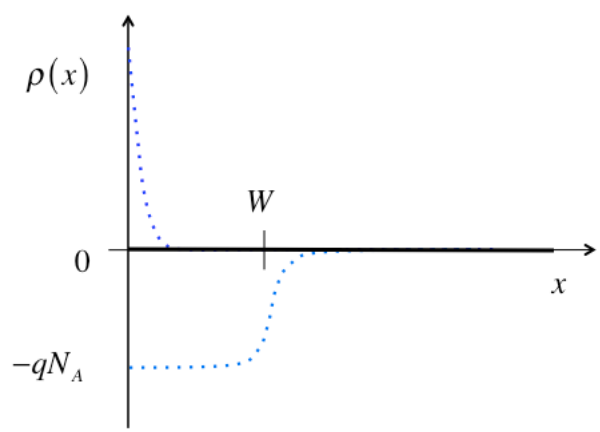
Holes pile up ("accumulate") near the surface, giving a large, positive space charge. Note that the band bending occurs in a much thinner region near the surface as compared to the case for depletion in 1b)(the dotted line).

1d) Assume $\phi_S = 0$ and sketch the energy band diagram and the charge density, $\rho(x)$ vs. position in the semiconductor.

Solution:



A zero surface potential means no band bending.

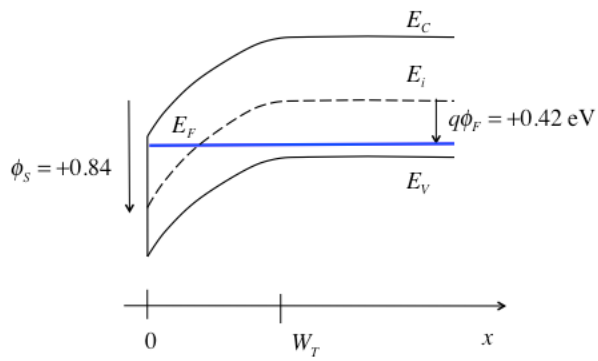


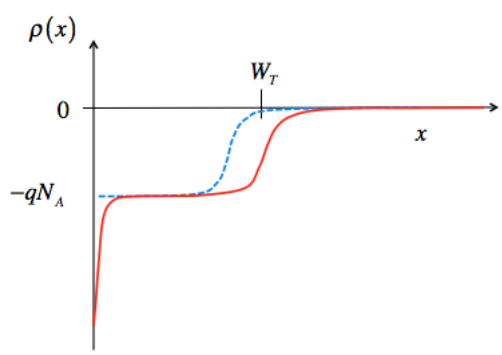
HW11 Solutions (continued):

No bandbending means no space charge in the semiconductor (the heavy line at $\rho(x)=0$).

1e) Assume $\phi_S = 2\phi_F$ and sketch the energy band diagram and the charge density, $\rho(x)$ vs. position in the semiconductor.

Solution:





As compared to 1b) (shown by the dotted line), the depletion layer is deeper, $W_T > W$, because the surface potential is larger. But now the intrinsic level at the surface is as far below the Fermi level as it was above the Fermi level in the bulk. So the concentration of electrons at the surface is the same as the concentration of holes in the bulk. This electron inversion charge piles up very close to the surface. (For analytical calculations, we treat it as a delta-function.)

HW11 Solutions (continued):

At $\phi_S = 2\phi_F$, an inversion layer is beginning to develop at the surface, but most of the charge is still in the depletion layer. As the surface potential increase just a little above $\phi_S = 2\phi_F$, the inversion charge builds up exponentially. The term "strong inversion" describes the situation for ϕ_S just a little above $2\phi_F$ where the inversion charge dominates.

To test your understanding, you should repeat this problem for an N-type semiconductor with $N_D = 10^{17}$ cm^{-3}. Note that ϕ_F is still defined as $\phi_F = \left(E_i - E_F\right)/q$, so ϕ_F is negative in an N-type semiconductor.

2) Get a feel for the magnitudes of important quantities by answering the following questions for a room temperature MOS capacitor with $N_A = 10^{18}$ cm^{-3} and an oxide thickness of 2 nm with $K_O = 3.9$.

2a) Compute $\phi_F = \left(E_i - E_F\right)/q$

Solution:

$$\phi_F = \frac{\left(E_i - E_F\right)}{q} = \frac{k_B T}{q}\ln\left(\frac{N_A}{n_i}\right) = 0.026\ln\left(\frac{10^{18}}{10^{10}}\right) = 0.479 \qquad \boxed{\phi_F = 0.479\text{ V}}$$

2b) Compute the depletion layer thickness, W, when $\phi_S = 2\phi_F$.

Solution:

$$W = \left[\frac{2K_S\varepsilon_0}{qN_A}\phi_S\right]^{1/2} = \left[\frac{2\times 11.8\times 8.854\times 10^{-14}}{1.6\times 10^{-19}\times 10^{18}}\times 2\times 0.479\right]^{1/2} = 3.54\times 10^{-6}\text{ cm}$$

$$\boxed{W = W_T = 3.54\times 10^{-6}\text{ cm} = 0.0354\ \mu\text{m} = 35.4\text{ nm}}$$

2c) Compute the electric field at the surface, $\mathcal{E}_S$, when $\phi_S = 2\phi_F$.

Solution:

We could use the formula: $\mathcal{E}_S = \left[\frac{2qN_A}{K_S\varepsilon_0}\phi_S\right]^{1/2}$

Or we could remember that:

HW11 Solutions (continued):

$$\phi_S = \frac{1}{2}\mathcal{E}_S W$$

Since we know the surface potential and W, the second way is easier.

$$\mathcal{E}_S = \frac{2\phi_S}{W} = \frac{2\times(2\times 0.479)}{3.54\times 10^{-6}} = 5.42\times 10^5 \text{ V/cm} \qquad \boxed{\mathcal{E}_S(2\phi_F) = 5.42\times 10^5 \text{ V/cm}}$$

It is important to note that although we are at the onset of inversion, we are assuming that most of the charge is still in the depletion region, so that we can still use depletion approximation equations.

2d) Compute the threshold voltage, V_T, assuming no metal-semiconductor workfunction difference. (This is the gate voltage needed to make $\phi_S = 2\phi_F$ and create an inversion layer in the semiconductor.

Solution:

According to SDF, eqn. (16.26)

$$V_G = \phi_S + \frac{K_S}{K_0} x_0 \mathcal{E}_S$$

If we define the oxide capacitance per cm^2 as

$$C_{ox} = \frac{K_S \varepsilon_0}{x_0}$$

and remember that the charge in the semiconductor, Q_S, is related to the electric field at the surface by

$$Q_S = -K_S \varepsilon_0 \mathcal{E}_S$$

then we can write the V_G expression as

$$V_G = \phi_S - \frac{Q_S}{C_{ox}}$$

which is easier to remember.

$$Q_S(2\phi_F) = -K_S\varepsilon_0\mathcal{E}_S(2\phi_F) = -11.8\times 8.845\times 10^{-14}\times 5.42\times 10^5 = -5.72\times 10^{-7} \text{ C/cm}^2$$

$$C_{ox} = \frac{K_O\varepsilon_0}{x_0} = \frac{3.9\times 8.854\times 10^{-14}}{2\times 10^{-7}} = 1.73\times 10^{-6} \text{ F/cm}^2$$

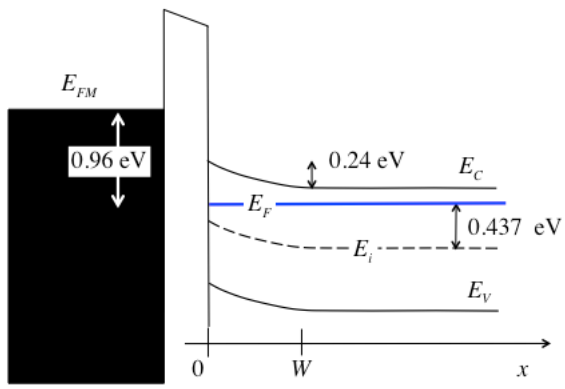
$$V_G(2\phi_F) = 2\phi_F - \frac{Q_S(2\phi_F)}{C_{ox}} = 0.958 + \frac{5.72\times 10^{-7}}{1.73\times 10^{-6}} = 1.29 \text{ V}$$

$$\boxed{V_G(2\phi_F) = V_T = 1.29 \text{ V}}$$

HW11 Solutions (continued):

Note that in practice, we would need to add a metal –semiconductor workfunction difference, which would shift the threshold voltage about 1 V negative and make V_T about 0.3 V.

3) Answer the following questions about the energy band diagram sketched below. Assume that the zero for electrostatic potential is in the semiconductor bulk, at large *x* and that there is no metal-semiconductor workfunction difference. Also assume that the relative dielectric constant of the oxide is $K_O = 3.9$.



3a) What it the surface potential, ϕ_S?

Solution:
It is the bandbending in the semiconductor. Since the bands bend up, the surface potential is negative. From the figure, we see:

$$\boxed{\phi_S = -0.24 \text{ V}}$$

3b) What gate voltage, V_G, is applied?

Solution:
The gate voltage is the difference between the Fermi level in the metal and the Fermi level in the semiconductor. Since the Fermi level in the metal is higher than the Fermi level in the semiconductor, the gate voltage is negative. From the figure, we see:

$$\boxed{V_G = -0.96 \text{ V}}$$

HW11 Solutions (continued):

3c) What is the voltage across the oxide, $\Delta\phi_{ox}$?

Solution:
Voltages add up according to Kirchoff's Law.

$$V_G = \Delta\phi_{ox} + \phi_S$$

$$\Delta\phi_{ox} = V_G - \phi_S = -0.96 + 0.24 = -0.72 \text{ V} \qquad \boxed{\Delta\phi_{ox} = -0.72 \text{ V}}$$

Note that $\Delta\phi_{ox} = \phi_{ox}\left(x = -x_0\right) - \phi_{ox}\left(x = 0\right)$; it is negative because the potential is more negative at the top of the oxide than at the oxide-Si interface.

3d) What is the doping density, N_D.

Solution:
We determine the carrier density in the bulk, which we assume is equal to the doping density, from

$$n_0 = n_i e^{\left(E_F - E_i\right)/k_B T} = N_D$$

Putting in numbers:

$$N_D = n_i e^{\left(E_F - E_i\right)/k_B T} = 10^{10} e^{0.437/0.026} = 1.99 \times 10^{17} \qquad \boxed{N_D = 1.99 \times 10^{17} \text{ cm}^{-3}}$$

3e) What is the width of the depletion region, W ?

Solution:

$$W = \left[\frac{2K_S\varepsilon_0}{qN_D}\left(-\phi_S\right)\right]^{1/2}$$

We know the surface potential from the figure, and we have just determined the doping density, so

$$W = \left[\frac{2K_S\varepsilon_0}{qN_D}\left(-\phi_S\right)\right]^{1/2} = \left[\frac{2 \times 11.8 \times 8.854 \times 10^{-14}}{1.6 \times 10^{-19} \times 1.99 \times 10^{17}} \times 0.24\right]^{1/2} = 3.97 \times 10^{-6} \text{ cm}$$

$$\boxed{W = 3.97 \times 10^{-6} \text{ cm}}$$

3f) What is the thickness of the oxide?

Solution:
The volt drop across the oxide is the electric field in the oxide times the thickness of the oxide:

HW11 Solutions (continued):

$$\Delta\phi_{ox} = \mathcal{E}_{ox} x_o \qquad\qquad x_o = \frac{\Delta\phi_{ox}}{\mathcal{E}_{ox}}$$

We deduced $\Delta\phi_{ox}$ in part 3c).

The electric field in the oxide can be determined from Gauss's Law (assuming no charge at the oxide-semiconductor interface.

$$K_O \varepsilon_0 \mathcal{E}_{ox} = K_S \varepsilon_0 \mathcal{E}_S \qquad\qquad \mathcal{E}_{ox} = \frac{K_S}{K_O}\mathcal{E}_S = \frac{11.8}{3.9}\mathcal{E}_S = 3.03\mathcal{E}_S$$

$$\mathcal{E}_S = -\left[\frac{2qN_D}{K_S\varepsilon_0}\left(-\phi_S\right)\right]^{1/2}$$

$N_D = 1.99\times10^{17}$ cm^{-3} (from part 3d)

$\phi_S = -0.24$ V (from part 3a)

Putting in numbers:

$$\mathcal{E}_S = -\left[\frac{2qN_D}{K_S\varepsilon_0}\left(-\phi_S\right)\right]^{1/2} = -1.21\times10^5 \text{ V/cm}$$

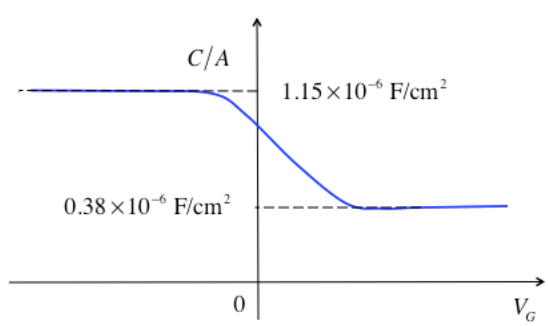
So

$$\mathcal{E}_{ox} = 3.03\mathcal{E}_S = 3.03\times -1.21\times10^5 = -3.66\times10^5$$

Finally,

$$x_o = \frac{\Delta\phi_{ox}}{\mathcal{E}_{ox}} = \frac{-0.72}{-3.66\times10^5} = 1.97\times10^{-6} \text{ cm} \qquad \boxed{x_o = 19.7 \text{ nm}}$$

4) A high-frequency MOS capacitance vs. voltage characteristic is sketched below. Answer the following question assuming that the semiconductor is silicon at room temperature.



HW11 Solutions (continued):

4a) Is the semiconductor N-type or P-type? Explain your answer.

Solution:

P-type. Because accumulation occurs for large negative voltages and inversion for large, positive gate voltages.

4b) What is the thickness of the oxide?

Solution:
In accumulation, the total capacitance is approximately the oxide capacitance.

$$C = C_{ox} = \frac{K_O \varepsilon_0}{x_o} = 1.15 \times 10^{-6} \text{ F/cm}^2$$

$$x_o = \frac{K_O \varepsilon_0}{C} = \frac{3.9 \times 8.854 \times 10^{-14}}{1.15 \times 10^{-6}} = 3.0 \times 10^{-7} \text{ cm}$$

$$\boxed{x_o = 3 \text{ nm}}$$

4c) What is the thickness of the depletion layer in inversion?

Solution:
We have 2 capacitors in series:

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_S}$$

$$C_S = \left(\frac{1}{C} - \frac{1}{C_{ox}} \right)^{-1} = \left(\frac{1}{0.38 \times 10^{-6}} - \frac{1}{1.15 \times 10^{-6}} \right)^{-1} = 0.57 \times 10^{-6} \text{ F/cm}^2$$

We also know that the semiconductor capacitance is related to the depletion layer thickness:

$$C_S = \frac{K_S \varepsilon_0}{W_T}$$

So

$$W_T = \frac{K_S \varepsilon_0}{C_S} = \frac{11.8 \times 8.854 \times 10^{-14}}{0.57 \times 10^{-7}} = 1.84 \times 10^{-6} \text{ cm} \qquad \boxed{W_T = 18.4 \text{ nm}}$$

HW11 Solutions (continued):

4d) Explain how you could calculate the doping density of the semiconductor.

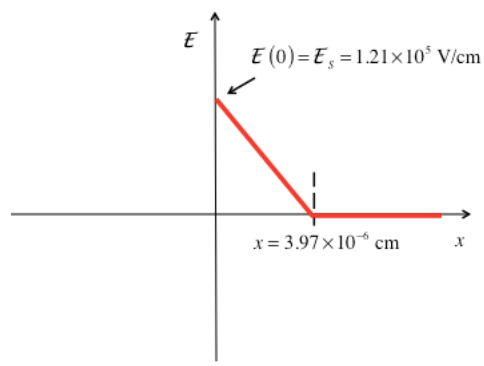
Solution:

$$W_T = \left[\frac{2K_S\varepsilon_0}{qN_A}\left(2\phi_F\right)\right]^{1/2} = \left[\frac{2K_S\varepsilon_0}{qN_A}\left(2\frac{k_BT}{q}\ln\left(\frac{N_A}{n_i}\right)\right)\right]^{1/2}$$

$$W_T = \left[\frac{K_S\varepsilon_0\left(k_BT/q\right)}{q}\right]^{1/2}\frac{\ln\left(N_A/n_i\right)}{\sqrt{N_A}}$$

Since W_T is known from the measurement, we can guess N_A and iteratively adjust it until the above equation is satisfied.

5) The problem concerns an MOS capacitor that consists of a metal electrode, a 2 nm thick silicon dioxide layer with $K_O = 3.9$, and a silicon substrate that begins at $x = 0$. The electric field in the semiconductor is shown below.



Answer the following questions.

5a) What is the surface potential, ϕ_S? (Assume that the potential is zero deep in the semiconductor.

Solution:

$$\phi_S = \frac{1}{2}\mathcal{E}_S W = 0.5\times\left(1.21\times10^5\right)\times\left(3.97\times10^{-6}\right) = 0.24$$

$$\boxed{\phi_S = +0.24\text{ V}}$$

The sign is positive, because the electric field > 0, so the potential at the surface much be greater than the potential in the bulk.

HW11 Solutions (continued):

5b)What is the doping density in the semiconductor?

Solution:

Use the Poisson equation:

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{K_S\varepsilon_0} = -\frac{qN_A}{K_S\varepsilon_0}$$

$$\frac{d\mathcal{E}}{dx} = \frac{0-\mathcal{E}_S}{W} = -\frac{\mathcal{E}_S}{W}$$

Putting the above two equations together, we fid

$$N_A = \frac{K_S\varepsilon_0}{qW}\mathcal{E}_S$$

$$N_A = \frac{11.8\times 8.854\times 10^{-14}}{\left(1.6\times 10^{-19}\right)\left(3.97\times 10^{-6}\right)}1.21\times 10^5 = 1.99\times 10^{17}$$

$$\boxed{N_A = 1.99\times 10^{17}\ \text{cm}^{-3}}$$

5c) What is the electric field in the oxide?

Solution:

The D-field is continuous:

$$K_O\varepsilon_0\mathcal{E}_{ox} = K_S\varepsilon_0\mathcal{E}_S \qquad \mathcal{E}_{ox} = \frac{K_S}{K_O}\mathcal{E}_S = \frac{11.8}{3.9}\mathcal{E}_S = 3.03\mathcal{E}_S$$

$$\mathcal{E}_{ox} = 3.03\times 1.21\times 10^5 = 3.67\times 10^5$$

$$\boxed{\mathcal{E}_{ox} = 3.67\times 10^5\ \text{V/cm}}$$

5d) What is the voltage drop across the oxide?

Solution:

$$\Delta\phi_{ox} = \mathcal{E}_{ox}x_o$$

$$\Delta\phi_{ox} = \left(3.67\times 10^5\right)\times 2\times 10^{-7} = 0.073$$

$$\boxed{\Delta\phi_{ox} = 0.073\ \text{V}}$$

HW11 Solutions (continued):

5e) What is the electrostatic potential in the metal gate?

Solution:

Add the volt drop across the semiconductor and the volt drop across the oxide:

$$\phi_{metal} = \Delta\phi_{ox} + \phi_S = 0.073 + 0.24 = 0.31\ \text{V}$$

$$\boxed{\phi_{metal} = 0.31\ \text{V}}$$